AYUSH JAYESHBHAI HALPATI

FULL-STACK DEVELOPER | GENAI & RAG INTEGRATION

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LinkedIn | GitHub | Portfolio

PROFESSIONAL SUMMARY

Computer Engineering graduate (2026) focused on building full-stack web applications with React/Next.js, TypeScript, APIs and PostgreSQL/Supabase. Extends web products with GenAI capabilities including LLM API integration, RAG pipelines, embeddings and retrieval. Built role-based operational workflows, database-enforced order claiming, realtime notifications and LLM-powered applications; also brings hands-on computer-vision model training and deployment experience.

TECHNICAL SKILLS

Frontend & Web: JavaScript, TypeScript, React.js, Next.js, HTML, CSS, responsive web UI

Backend & Data: Node.js, Express.js, REST APIs, SQL, PostgreSQL, MySQL, MongoDB, Supabase Auth/Realtime/RLS/RPCs, Git/GitHub, Vercel

GenAI & RAG: Groq API, Gemini API, Ollama, RAG, prompt engineering, embeddings, semantic retrieval, LangChain, ChromaDB, Sentence Transformers

Python & ML: Python, PyTorch, YOLOv8, OpenCV, SAM, DINOv2, K-Means, scikit-learn, NumPy, Roboflow, dataset curation/augmentation

SELECTED PROJECTS

TableFlow | Next.js, TypeScript, Supabase, PostgreSQL, Gemini | GitHub / Live

- Solo-built for VibeAthon 6.0 and extended over the following month: staff roles, Row-Level Security, realtime notifications, Web Push and Gemini-powered demand forecasting.
- Implemented database-enforced exclusive order claiming with row-locking PostgreSQL RPCs and verified behaviour against simultaneous claim attempts.

DeliveryProof | Next.js, Supabase, PostgreSQL, Groq API | GitHub / Live demo / Video / Devpost

- Solo-built for the AI Builders Hackathon 2026: implemented Company, Driver and Reviewer workflows around a chronological delivery-evidence timeline; AI summaries use structured events while timestamps and state transitions remain deterministic.

ZENI - Private AI Thinking Partner | React, Vite, Ollama, Groq | GitHub / Live

- Built an LLM workspace with project-context-aware prompts, streaming responses, structured-output parsing and prompt compilation from multi-turn sessions; supports local Ollama and cloud Groq modes.

NCERT AI Tutor | Python, LangChain, ChromaDB, Groq, Streamlit | GitHub / Live

- Built a RAG pipeline over Class 10 NCERT Maths and Science PDFs using chunking, local Sentence-Transformer embeddings, subject-filtered ChromaDB retrieval and Groq (Llama 3.1) generation in English and Hindi; added layered fallback handling for out-of-syllabus questions.

EXPERIENCE

AI/ML Intern (Full-Semester External Project) - Invisible Fiction | 15 Dec 2025 - 10 Apr 2026

V.V. Nagar, Anand, Gujarat | Project repository / Demo video

- Trained a YOLOv8 Nano segmentation model for 25 mechanical-part classes on a 6,000-image dataset (real images, Blender CAD renders and background-replacement composites).
- Diagnosed background overfitting and applied background-replacement augmentation, reducing false positives by 30%+; built a data pipeline with Roboflow labels, SAM-assisted masks, manual review and DINOv2/K-Means near-duplicate filtering.
- Integrated the lightweight model into an Android app using Unity/AR Foundation with stability-gated inference and General/Specific detection modes.

ACHIEVEMENT & EDUCATION

VibeAthon 6.0 (NXTGENSEC, 25-27 Jul 2026): Certificate of Achievement; Top 60 of 800+ submissions for TableFlow.

B.Tech, Computer Engineering - Birla Vishvakarma Mahavidyalaya (BVM) Engineering College, Anand, Gujarat | 2022-2026